Infineon Powerful solution for Humanoid Robot

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Feb 2026

restricted

Humanoid Robot is one of the focus applications in Robotics

Industrial Robots

Heavy industrial automation

INOVANCE 汇川机器人

ESTUN ROBOTICS

- New to C3
- Mature market
- 6-axis & 4-axis Robots

- 300K units sales in 2024, -4.5% YoY
- 163K units sales in 1H 2025, +16% YoY
- Cobot share in all industrial robot: >10%

Cobots

Collaboration & automation

Specific applications: surgery

JAKA 节卡

AUBO

- Flexible production
- Labor shortage
- Fast growing

Mobile Robots

Logistic & warehouse

Hospitality Robot

Geek+

云迹科技 YUNJI TECHNOLOGY

- Logistics automation
- Last mile delivery
- Cost savings

- Huge investment to humanoid robot, forecast >12,000 units in 2025
- Various industrial & consumer scenarios for mobile robot
- GC leading RVC players contribute 57% global shipment in 1H 2025

Domestic Robots

Vacuum Robots

Lawn mower robots

+ Other domestic robots

ECOVACS ROBOTICS

roborock

- Everyday helping robots
- AI drives Humanoid Robot development
- Aging population & labor shortage

Humanoid Robots

Exoskeleton

Humanoid

+ other categories

Unitree Robotics

Fourier Intelligence

Drones

Consumer

Commercial

eVTOL

dji

美团 Meituan

AUTOFLIGHT

丰翼无人机 PHOENIX WINGS

- Video & social media content creations
- Agriculture, Construction/Traffic supervision
- Delivery Drones

- Stable growth for both consumer drone & commercial drone
- EHANG EH216-S got eVTOL license

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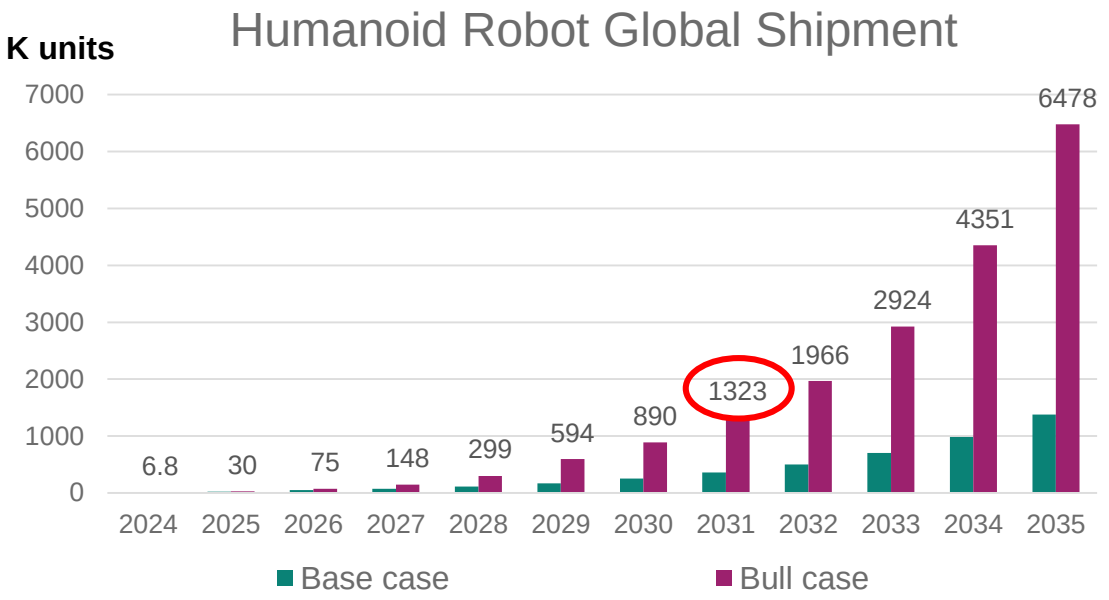
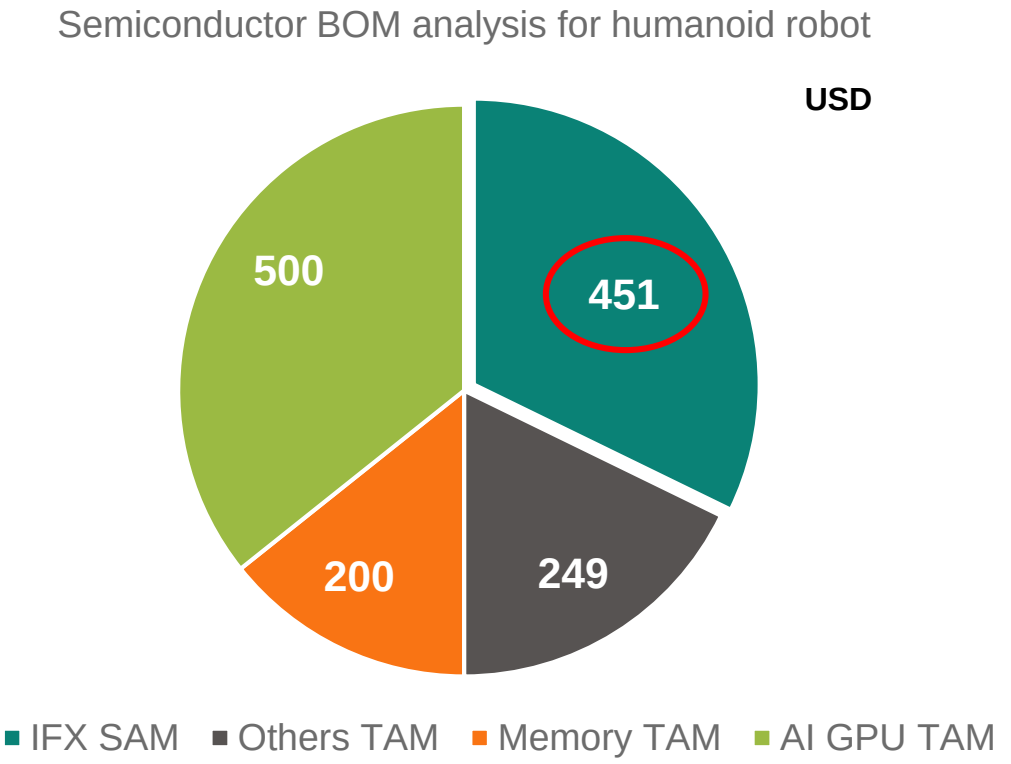
Humanoid Robot Market

Source: Goldman Sachs



2025-2035	>50%	-	Base case growth rate
CAGR	>70%	-	Bull case growth rate



- 1M units shipment milestone for Humanoid Robot in 2030-2031
- Market driver: Aging population, Increasing labor cost, AI extension to physical
- Humanoid Robot forecast in GC:
  - 15K units in 2025, ~200% YoY growth, accounting for ~70% global market share

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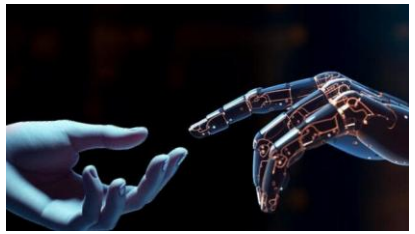
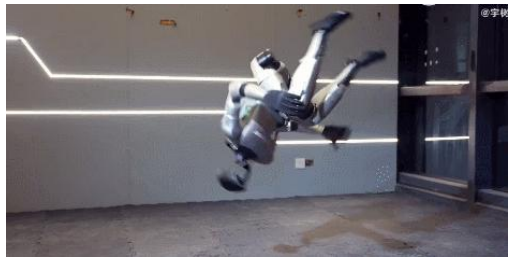
Market overview





Market View

- Most humanoid robot companies focus on motion control to make robot can run/walk smoothly (acceleration by AI)
- Fast move for new technology and innovation design in humanoid robot
- Different understanding about humanoid robot system from customers
- Humanoid robot is one of the target applications for GaN
- No standard or regulation for humanoid robot yet, while most players believe safety will be a key for large-scale deployments

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Humanoid Robot System Architecture

Power
Signal

Power sensor

DC-DC
48V → 12V/5V

Power AI

DC-DC
48V → 19V/10A

Battery 48V
- BMS controller
- MOSFET

Charger
- AC-DC controller
- MOSFET

Environmental sensor

- RGB Camera
- Accelerometers
- Depth Camera (iToF)
- Compass
- Microphone
- Gyroscopes
- Barometric sensor
- Torque sensor
- Radar
- Tactile sensor
- Lidar
- Temperature sensor

Decision system

GPU (NVIDIA)
AI processing

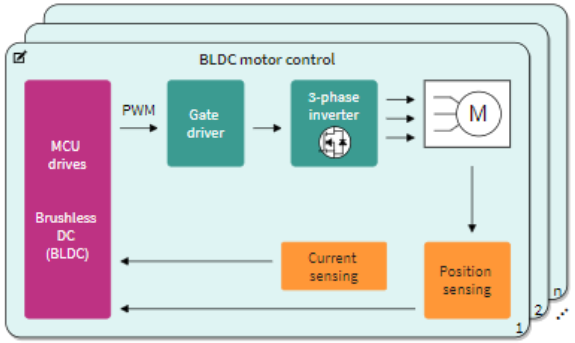
CPU or ARM
Motion control

EtherCAT or CAN

Main Compute boards are outsourcing with Wi-Fi & Multiphase controller opportunities

Actuator

- Arm
- Leg
- Waist / Neck
- Dexterous hand



- MCU & Gate Driver
- MOSFET & MV GaN
- Current sensor
- Position sensor

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Infineon product portfolio for every humanoid robot functional block

Battery charging

AC-DC conversion
Microcontroller:
XMC™ 1000, PSOC™ 4, PSOC™ Control C3P
Power MOSFETs/HEMTs:
OptiMOS™, CoolMOS™
CoolGaN™ & CoolSiC™
Gate driver: EiceDRIVER™

Wireless charging
Wireless charging IC
Authentication: OPTIGA™ Trust

Charging station position sensing
Hall switch, 3D magnetic sensor

Battery management

Battery management
Microcontroller: XMC™, PSOC™, AURIX™
Cell monitoring and balancing IC
Isolated transceiver IC
Current sensing: XENSIV™ Current Sensors

Protection MOSFETs
OptiMOS™ & OptiMOS™ Linear FET

Battery authentication
OPTIGA™ Authenticate S

Power management & distribution

Voltage conversion
LDO & Step-down switching regulator
Single and dual MOSFETs: OptiMOS™
Power management IC

Smart Power switches
PROFET™, SPOC™ and HITFET™

Small signal transistors
OptiMOS™

ESD Protection
Low capacitance and multi-purpose ESD diode

Charger

Wireless charging

BMS

Smart power

BLDC motor control

Motor control

Microcontroller
PSOC™ Control C3M
PSOC™ 4 & 6
AURIX™ TC3, TC4
MOTIX™ MCU

Power MOSFETs
OptiMOS™, StrongIRFET™ MOSFETs
CoolGaN™ HEMTs

Gate Driver & Integrations
EiceDRIVER™
MOTIX™ Driver

Magnetic Sensors
XENSIV™: Position and Current Sensors

Artificial intelligence

HMI

Main and HMI control

Microcontroller
XMC™ 4000, XMC™ 7000
PSOC™ 6
PSOC™ Edge
AURIX™

HMI controller
PSOC™ 4 PSOC™ 6 with CAPSENSE™

CAN transceiver
TLE9251VLE, TLE9351VSJ

USB – Serial bridge
EZ-USB™

Memory
SEMPER™ NOR flash
EXCELON™ F-RAM

Sensing

Wireless communication

Security

Navigation and HMI sensing

XENSIV™ 24/60 GHz radar
Obstacles and gesture detection, materials differentiation, speed over ground

REAL3™ 3D IR ToF sensor
Objects recognition, obstacles avoidance, navigation and tracking
hToF Solution

XENSIV™ MEMS microphone
Sound detection / HMI, objects detection and tracking

XENSIV™ Air pressure sensor
Height and touch sensing

XENSIV™ 3D magnetic sensor
HMI joystick, jog-wheels, collision detection

XENSIV™ hall latch/switch
Collision detection

CAPSENSE™ capacitive sensing controller
Rain detection

Connectivity and security

Ethernet
BRIGHTLANE™ PHY and switch

Memory
SEMPER™ NOR flash

Wi-Fi & Bluetooth® / BLE
Connected MCUs: PSOC™ 4 BLE, PSOC™ 6 BLE, BT®/BLE SoC and modules
Combos: AIROC™

Embedded-SIM Connectivity
OPTIGA™ Connect IoT

Security
OPTIGA™ Trust M, TPM

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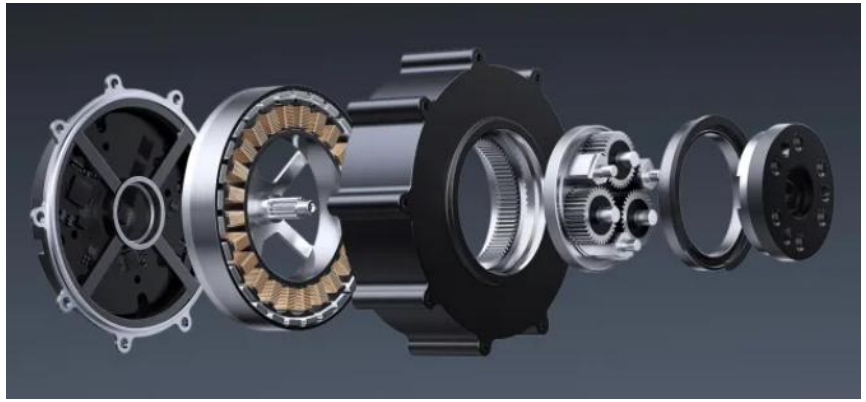
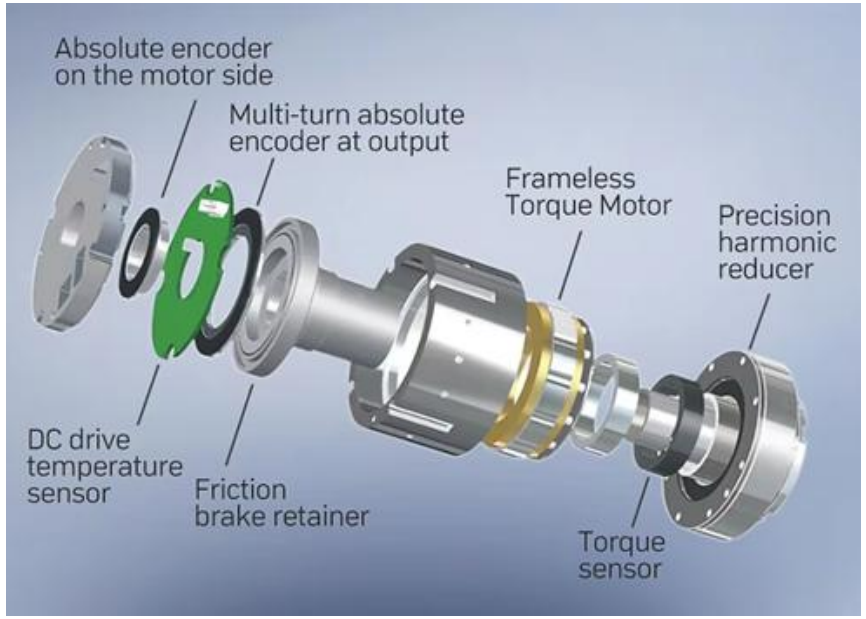
Focus functional block (1) - Joint module (motor control)





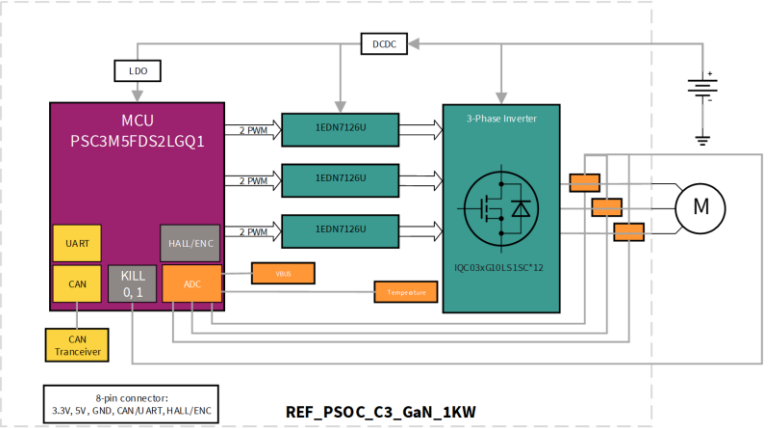
Spec & Requirement

- **Motor & Driver**: hollow shaft motor is popular with circular driver board
- **Reducer**: harmonic drive (100:1) in arm, planetary reducer (10:1) in leg
- **Encoder**: two encoders to detect motor and reducer's speed & position
- **Challenge:**
  - Limited space with many components inside
  - Peak power requirement: 3 times (rated power) & 3S
  - >30 joint modules per robot, need to control joint module's weight

Strategy

Infineon total solution including:

- MCU: PSoC C3M5 *1
- MV GaN: IGC016K10S2 *6/12
- Gate Driver: 1EDN7126U *6
- Current sensor: TLI5570/1/2 *3
- Position sensor: TLI49012 *1



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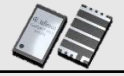
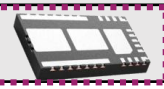
Infineon CoolGaN™ LV/MV product portfolio

Sales Code	Voltage class	$R_{DS(on),typ}$	Package type (mm)	Package figure	MP time	Positioning
IGC033S101	100 V	2.4 mΩ	PQFN 3x5 BSC		Production	p2p compatible with Navitas and package size compatible with EPC and Innoscience
IGC033S10S1			PQFN 3x5 DSC		Production	
IGB110S101		9.4 mΩ	PQFN 3x3 BSC		Production	
IGB110S10S1			PQFN 3x3 DSC		Production	
IGB070S10S1		5 mΩ	PQFN 3x3 DSC		Production	p2p compatible with standard Si products. Innoscience also recently announced plans for 100V GaN in same footprints
IGD015S10S1		1.1mΩ	RQFN 5x6 DD		Q3'25 EES Q1'25	
IGC019S06S1	60 V	1.3 mΩ	PQFN 3x5 DSC		Production	
IGC025S08S1	80 V	1.8 mΩ	PQFN 3x5 DSC		Production	
IGE033S08S1		2.3 mΩ	RQFN 3.3x3.3 SD		Q3'25 EES Q1'25	
IGC037S12S1	120 V	2.7 mΩ	PQFN 3x5 DSC		Production	
IGC090S20S1	200 V	6.7 mΩ	PQFN 3x5 DSC		Q1'25 EES available	
IGK080B041S	40 V BDS	6 mΩ	WLCSP 1.7x1.7		Q1'25 ES available	p2p compatible with Innoscience
IGK120B041S	40 V BDS	9 mΩ	WLCSP 1.2x1.7		Q2'25 EES available	
IGK048B041S	40 V BDS	4 mΩ	WLCSP 2.1x2.1		Q2'25 EES available	
IGJ024H10S2	100 V Drive	2.4 mΩ (x2)	RQFN 4.5 x 7		Q1'26 EES in Q2'25	p2p compatible with TI

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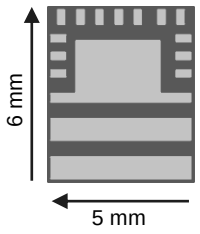
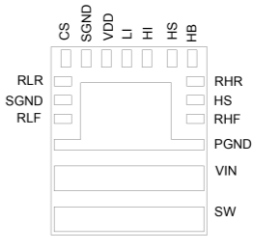
Half-Bridge Integrated Power Stage
G5 CoolGaN™ w/ Driver, Current Sense and Protections

G5 CoolGaN 60V | 80V | 100V, in RQFN 7x4.5 and 5x6 Package

RQFN 5x6 DSC

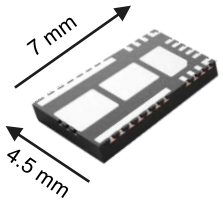


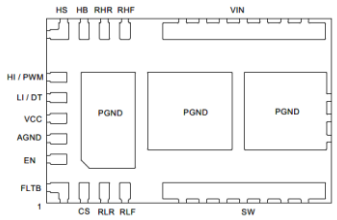


Voltage	$R_{DS(on)}$ typ.	Sales Code
60V	1.5 mΩ	IGD015H06Sx2
80V	1.7 mΩ	IGD017H08Sx2
100V	2.2 mΩ	IGD022H10Sx2

Target DS available

RQFN 7x4.5 DSC





Voltage	$R_{DS(on)}$ typ.	Sales Code
60V	1.5 mΩ	IGJ015H06Sx2
80V	1.7 mΩ	IGJ017H08Sx2
100V	2.2 mΩ	IGJ022H10Sx2

Target DS available

Features

- LS and HS **bidirectional current sense** +/- 3% (typ) accuracy
- Integrated **protections** (OCP, SCP, OTP)
- Suitable for up to **3 MHz** switching
- Adjustable drive strength
- Dual side cooled **RQFN pkg. in 2 variants**, **5x6 and 7x4.5**
- Fast propagation delay < 20ns | Minimum pulse width 10ns
- Optional **Interlock** feature

Benefits

- **Ease of use**: Integrated driver & current sense, optimized gate and power loops
- **Safe operation**: Miller clamp and bootstrap circuit, OCP, SCP, OTP
- **High power density and compact design**: Small, low-profile package in established footprints and dual side cooling
- **High efficciency**: Low conduction & switching losses ($R_{DS(on)}$, Q_{oss}, Q_G)

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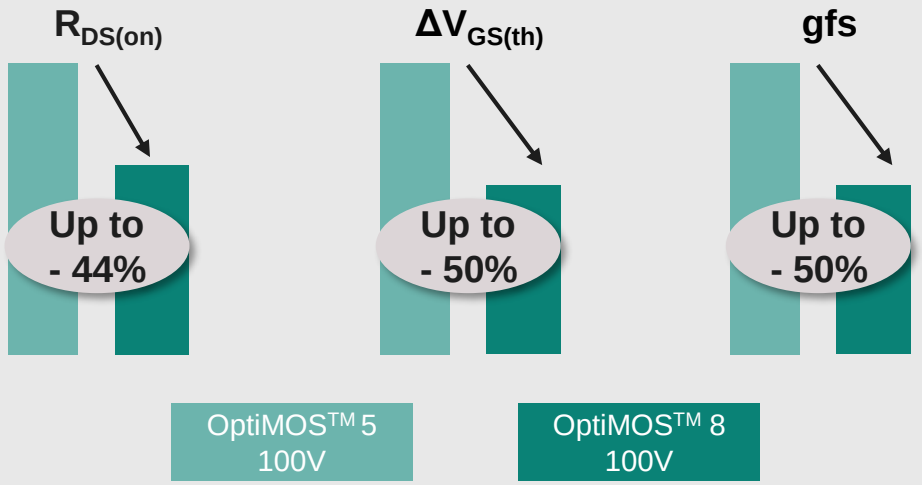
OptiMOS™ 8 Power MOSFETs in 100 V
Revolutionizing drives and battery protection applications

Value proposition (vs. OptiMOS™ 5)

- **Lower conduction losses** due to **lower $R_{DS(on)}$ x A**
  - Higher peak (2min) & steady state (60min) current capability
  - Enabling potential use of FR-4 (TOLL) instead of Al-IMS (D2PAK-7P)
- **Improved paralleling capability** due to **low (<1V) $V_{GS(th)}$** spread and **low transconductance (gfs)**
- **Excellent EMI behavior** due to **soft body diode**
- **Easy design transition** without major layout adaptation



Portfolio and timelines subject to change

EES: Available ES: Q3/CY25 Release: Q4/CY25

Pck	Product Name	Target R_{Dson} max. @10V	Category	Application D...Drives B...Battery protection
SSO8 5x6	**ISC019N10NM8**	1.94	BiC	D+B
	ISC033N10NM8	**3.3**	Price/Performance vs. BSC035N10NS5	**D**
	ISC040N10NM8	4.0	Price/Performance vs. BSC040N10NS5	D
TOLL w/ Cu-Clip	IPTH83N10NM8S	0.83	BiC	D+B
TOLL	**IPT009N10NM8**	0.92	BiC	D+B
	IPT014N10NM8	1.36	Price/Performance vs. IPT015N10N5	D
	IPT018N10NM8	1.86	Price/Performance vs. IPT020N10N5	D
D²PAK 7-pin	IPF009N10NM8	0.96	BiC	D+B
	IPF014N10NM8	1.43	Price/Performance vs. IPB017N10N5	D
	IPF019N10NM8	1.95	Price/Performance vs. IPB024N10N5 IPB025N10N3	D
D²PAK 3-pin	IPB013N10NM8	1.36	BiC	D+B
	IPB018N10NM8	1.78	Price/Performance vs. IPB020N10N5	D

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Infineon packaging advancements - 1

Industry standard package

Normal and logic-level gate drive

High current carrying capability

D^{2}PAK

10.31 x 15.88 x 4.57 mm

Industry standard package

Normal and logic-level gate drive

Up to **23%* increased current rating** vs D^{2}PAK

D^{2}PAK 7pin

10.31 x 15.88 x 4.57 mm

– Industry standard package

– **60% space reduction** vs D^{2}PAK 7pin

– Up to **63%* increased current rating** vs D^{2}PAK 7pin

TO-Leadless (TOLL)

10.10 x 11.88 x 2.40 mm

TOLG

– Optimized for TCoB performance

TOLT

– Optimized for thermal and TCoB performance

TOLG - TOLL w/ Gullwing leads

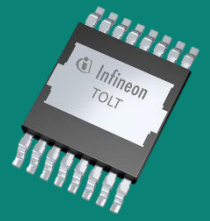
10.10 x 11.90 x 2.40 mm

– Compatible footprint to TOLL

– **Excellent solder joint reliability** on Al-IMS boards

– Up to **57%* increased current rating** vs D^{2}PAK 7pin

TOLT



10.10 x 15.20 x 2.35 mm

– **Top side cooling** package addressing **high current** applications

– Up to **52%* increased current rating** vs D^{2}PAK 7pin

* dependent on breakdown voltage of the MOSFET

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Infineon packaging advancements - 2

SuperSO8 5x6 (SSO8)

5.35 x 6.10 x 1.20 mm

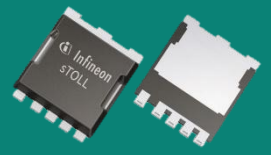
- Industry standard 5x6 mm package
- **Fused source leads** for improved current carrying capability
- High current rating

sTOLL

High current carrying capability which is ideal for surviving locked rotor events in BLDC motor applications.

Leadless package with low package resistance and minimized stray inductance.

sTOLL shrink TOLL



7.20 x 8.20 x 2.40 mm

- Bridges the gap between SSO8 and TOLL
- Infineon's solution to onsemi's 8x8 package
- **~55%*** current rating improvement vs SSO8

SuperSO8 Source-Down

Improved R_{thJA} , R_{thJC} for better transfer of power losses

Center-Gate option enables optimized parallelization

Scalable Power Block

Significantly improved form factor between 40% - 60% compared to standard SSO8 footprint

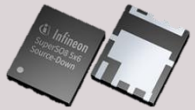
Dual-Side Cooling variant on roadmap for improved thermals

PQFN 8x6

Significantly improved form factor by ~60% compared to similarly specified TOLL device

Dual-Side Cooling variant on roadmap for improved thermals

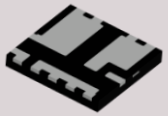
SuperSO8 Source-Down



5.05 x 6.05 x 0.75 mm

- Allows for higher Power Density in same footprint
- Same PCB routing
- **~30%*** current rating improvement vs SSO8

Scalable Power Block



6 x 6.3 mm

- High chip to package ratio for improved form factor
- Power density increased by **~50%** vs. 2xSSO8
- Simplicity in layout via half-bridge configuration

PQFN 8x6

8 x 6 mm

- Best in-class chip to package ratio
- Power density increased by **~25%** vs. PQFN 8x8
- Same pin placement as PQFN 5x6 for scalability

* dependent on breakdown voltage, $R_{DS(on)}$ of the MOSFET

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Infineon packaging advancements - 3

SuperSO8 5x6 (SSO8)

5.35 x 6.10 x 1.20 mm

- Industry standard 5x6 mm package
- **Fused source leads** for improved current carrying capability
- High current rating

SSO8 Dual-Side Cooling

- 41% volume reduction vs SSO8 for improved power density
- 95% improvement in $R_{thJC(top)}$ vs SSO8 for improved thermals

SuperSO8 Dual-Side Cooling (DSC)

5.05 x 6.05 x 0.75 mm

- Same footprint as SuperSO8 5x6
- **Exposed top** for dual-side cooling, improved thermals
- Similar current rating to SSO8

SuperSO8 Source-Down DSC

- Improved $R_{thJC\text{-}top}$ for better transfer of power losses
- Center-Gate option enables optimized parallelization

SuperSO8 SD DSC

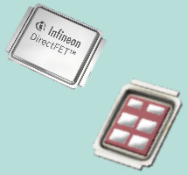
5.05 x 6.05 x 0.75 mm

- Same footprint as SuperSO8 5x6
- **Exposed top** for dual-side cooling, improved thermals
- ~30%* current rating improvement vs SSO8

DirectFET™ Mcan

- 43% volume reduction vs SSO8 for improved power density
- 90% improvement in $R_{thJC(top)}$ vs SSO8 for improved thermals
- RoHS 6/6 compliant

DirectFET™ Medium Can



5.05 x 6.35 x 0.70 mm

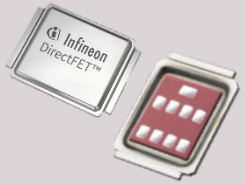
- **Source-down** for improved thermals thru the PCB
- **Exposed top** for dual-side cooling, improved thermals
- RoHS 6/6 compliant

* dependent on breakdown voltage of the MOSFET

DirectFET™ Lcan

- Limited to StrongIRFET™ silicon technology
- Can replace 2x Mcan DirectFET™, SuperSO8, SuperSO8 Super Cool devices in many apps.

DirectFET™ Large Can



7.10 x 9.15 x 0.74 mm

- **Source-down** for improved thermal thru the PCB
- **Exposed top** for dual-side cooling, improved thermals
- Similar current rating to SSO8

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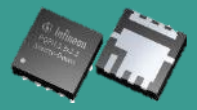
Infineon packaging advancements - 4

PQFN 3.3x3.3 SD Center-Gate

Optimized for paralleling due to center-gate pin placement and increased Creepage between source and drain

0.70 mm creepage distance vs 0.54 mm for standard Source-Down

PQFN 3.3x3.3 Source-Down (SD)



3.40 x 3.40 x 1.10 mm

Source-down for the next level of power density

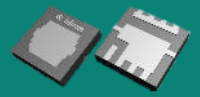
Same PCB routing

Increased fused lead area

increased I_{max}

~31%* current rating improvement vs PQFN 3.3x3.3

PQFN 3.3x3.3 SD DSC



3.40 x 3.40 x 0.75 mm

Similar benefits of PQFN 3.3x3.3 Source-Down (SD)

Increased power density and enhanced thermals due to dual-side cooling

Lower package profile

DSC = Dual-side cooling

Industry standard package

Normal and logic-level gate drive

High current rating

PQFN 3.3x3.3

3.40 x 3.40 x 1.10 mm

PQFN 3.3x3.3 Fused source leads

3.40 x 3.40 x 1.10 mm

Industry standard package

Fused source leads for improved current carrying capability

~9%* current rating improvement vs PQFN 3.3x3.3

* dependent on breakdown voltage of the MOSFET

PQFN 3.3x3.3 SD

25% improvement in $R_{DS(on)}$ and 22% reduction in R_{thJC} » increased power density

Source-down allows for improved die utilization and thermal capabilities

Large copper clip for improved current handling and thermals

PQFN 3.3x3.3 SD Center-Gate

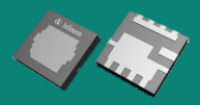
3.40 x 3.40 x 1.10 mm

Same benefits as PQFN 3.3x3.3 SD

Center-gate optimized for parallelization of MOSFETs

~31%* current rating improvement vs PQFN 3.3x3.3

PQFN 3.3x3.3 SD Center-Gate DSC



3.40 x 3.40 x 0.75 mm

Similar benefits of PQFN 3.3x3.3 SD Center-Gate

Increased power density and enhanced thermals due to dual-side cooling

Lower package profile

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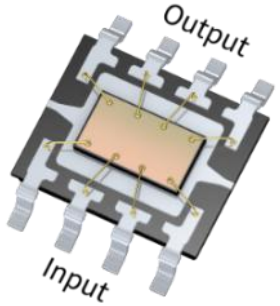
EiceDRIVER™ gate driver IC technology overview
Wide portfolio to best fit with application requirements

Non-isolated GD

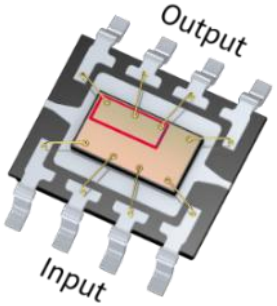
LoS Low-side



- Monolithic construction of ground-reference gate drivers for 20 to 25 V supply voltage applications
- **Comprehensive families** of single- and dual-low-side drivers with flexible options for output current, logic configurations and UVLOs (plus non-isolated TDI)
- Uses **rugged and high-performance technologies** of HVIC process or the latest **state-of-the-art 130-nm** process
- **Industry-standard** DSO-8 and small form-factor SOT23, WSON and TSNP packages

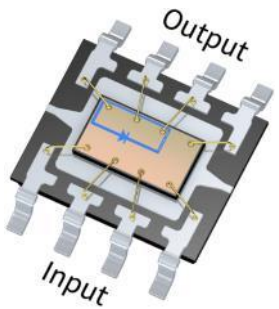
Level-shift GD

JI Junction isolation (JI)



- Monolithic construction of 1 to 6 gate drive channels up to 1200 V rating
- Industrial pioneering high-voltage IC **(HVIC) technology** used in all high-voltage gate drive applications
- Optional Integrated bootstrap- FET circuit (200 Ω typ.)
- **Negative transient immunity** to prevent latch-up: -40 V for 100 ns
- Common mode transient immunity (CMTI): 50 V/ns, typ.
- **Gen 2** technology (**IR** prefix): Industrial pioneering HVIC process
- **Gen 5** technology (**IRS** prefix): Cost-effective pin-to-pin versions of Gen 2

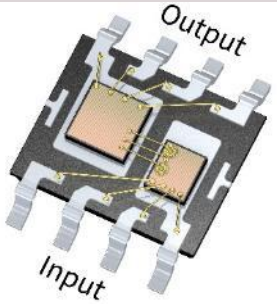
SOI Silicon on insulator (SOI)



- Monolithic construction of 2 to 6 gate drive channels 600, 700V, 1200 V rating
- **Infineon SOI technology** for high-voltage applications
- Built-in **PN-based bootstrap diode** (36 Ω typ.) for simplified bootstrap operation & reduced PCB area
- **Negative transient immunity** to prevent latch-up: -100 V for 300 ns
- Common mode transient immunity (CMTI): 50 V/ns, typ
- **>50% lower level-shift losses** allowing for higher efficiency, higher frequency operation, smaller heat sinks, and higher reliability

Isolated GD

CT Coreless transformer



- Two separate chips solution with **magnetic coupling** providing **galvanically isolated** single- and dual channel gate drivers
- **VDE 0884-11 isolation technology** providing **isolation up to 8 kV$_{pk}$ V$_{IOTM}$** and up to **±2300 V functional isolation**
- Common-mode transient immunity (**CMTI**) of **more than 300 V/ns**
- **Strongest gate-drive output currents** (up to ±18 A) reducing need for external booster circuits
- **Reliable protection options** like DESAT, Miller clamp, Analog/Digital configurability

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EiceDRIVER™ gate drivers are offered for a wide variety of configurations and voltage ranges

Gate Driver Configuration			25 V	25 - 100 V	160 V	200 V	500 V	600 V	650 V	1200 V	2300 V
Gate Drivers	1-Channel	Isolated								CT	CT
		High-Side		NIS JI		NIS JI	JI	JI	SOI		
		Low-Side	NIS						CT	CT	
	2-Channel	Isolated				CT					
		High-Side		NIS							
		Low-Side	NIS								
		High and Low-Side			SOI	JI	JI	JI SOI	SOI	JI SOI	
		Half-Bridge		JI	SOI	JI		JI SOI	SOI	JI SOI	
	4-Channel	H Bridge		JI				JI			
	6-Channel	Three-Phase Bridge			SOI	SOI		JI SOI	SOI	JI SOI	

NIS Non-Isolated (N-ISO) JI Junction Isolation (JI) SOI Silicon on Insulator (SOI) CT Coreless Transformer (CT)

Note: Voltage class on the top row is defined based on different driver configurations for the Max Voltage class. 1. For high-side, high-and low-side, half bridge and three phase gate drivers, voltage class is defined as switch break down voltage in applications. 2. For low side drivers (N-ISO), voltage class is defined as maximum operating range supply voltage. 3. For special cases as 1EDN-TDI (N-ISO), voltage class is defined as maximum bus voltage (highest floating voltage it can manage).

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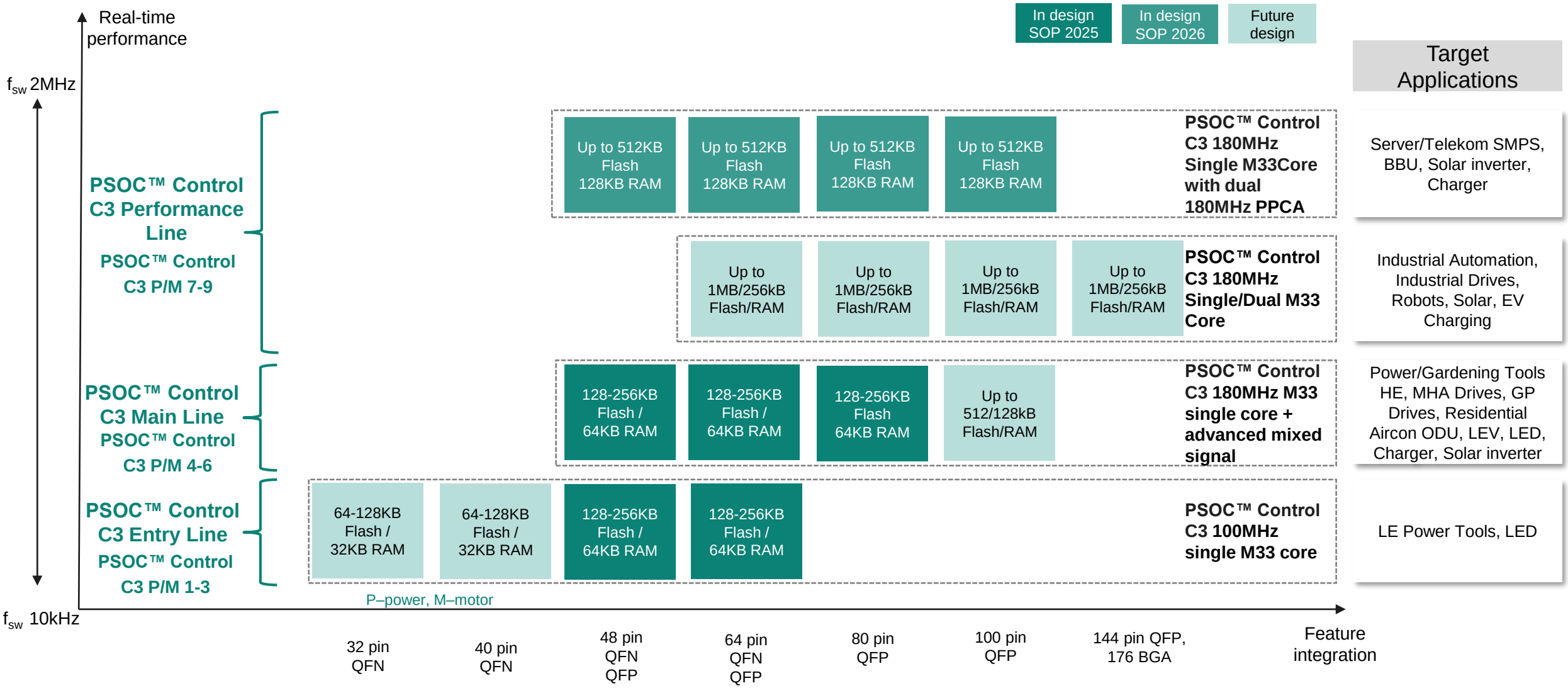
PSOC™ Control C3 Product Portfolio



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BRIGHTLANE™ PHY roadmap

Product Family	Production	<2025	2025 Q1	2025 Q2	2025 Q3	2025 Q4	2026 Q1	2026 Q2	2026 Q3	2026 Q4	2027 Q1	2027 Q2	2027 Q3	2027 Q4	Concepts
100 Mb	Infineon Ash Q1200/1 – 100BASE-T1 w/ TC10 & MACsec Infineon Q1010 – 100BASE-T1 Infineon Birch Q111x – 100BASE-T1 PHY w/ TC10								Infineon PineCone FE Q128x	• Octal 10/100B-T1 Port Extender					
1 Gb	Infineon Ash Q2220/1 – 1000BASE-T1 w/ TC10 & MACsec Infineon Sequoia Q211x – 100/1000B-T1 Infineon EA1512 – RGMII to SGMII						Infineon PineCone GE Q2240	• Quad 100/1000B-T1 Port Extender							
Multi-gig		Infineon Acacia Q324x	• 2.5/5/10GBASE-T1 PHY with MACsec												Infineon PineCone ME – • Quad 2.5Gb-T1 Port Extender Infineon Sal – • 2.5 Gb -T1 PHY Infineon Cassia – • 10 G/25Gb -T1 PHY

Sampling — In Development — In Planning Phase — Production

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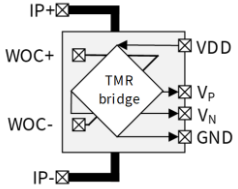
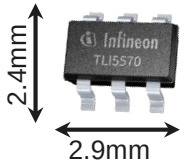
TLI557x series TMR current sensor

Preliminary data, subject to change during development

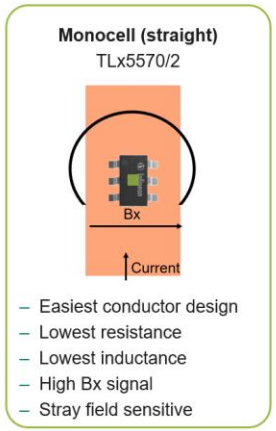
Description

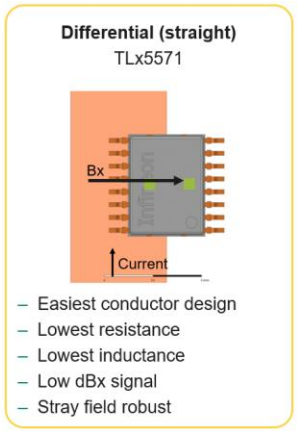
TLI5570/2 in SOT23 6pin package is a cost-competitive TMR based current sensor as Shunt replacement (non-amplified)





TLI5571 has two sensors in one chip to achieve differential detection





Features (preliminary)

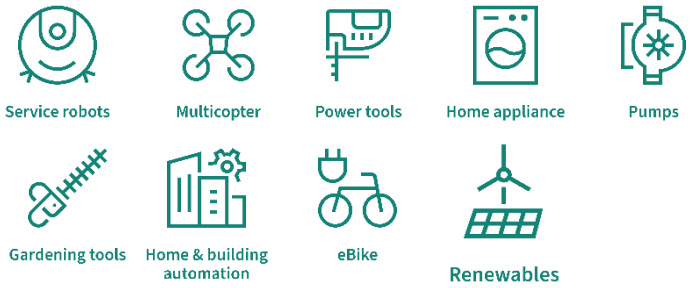
- Optimal for high currents in the range 30A - 2kA (can measure <30A, albeit with reduced accuracy)
- Miniature SMD package (2.4mm x 2.9mm)
- High bandwidth (**~500MHz**) and high accuracy (**3%-5%**)[1] & SNR (temperature trimming to be performed by the customer for achieving this accuracy)
- Discrete bridge variant with two outputs
- Contactless, galvanically isolated current measurement for reduced power losses and parasitic inductance
- Operational temperature up to **+125°C**

1) Accuracy range for temperatures -20C to 80C

Advantages & Applications

- Superior to shunt solutions due to additional features (no power loss in system, galvanic isolation, reduced BOM, smaller PCB footprint) and improved accuracy
- Cost-competitiveness on-par with existing shunt solutions
- Very few comparable products in market (from competitors)

Applications



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TLI49012 - Digital Angle Sensor with self-calibration feature

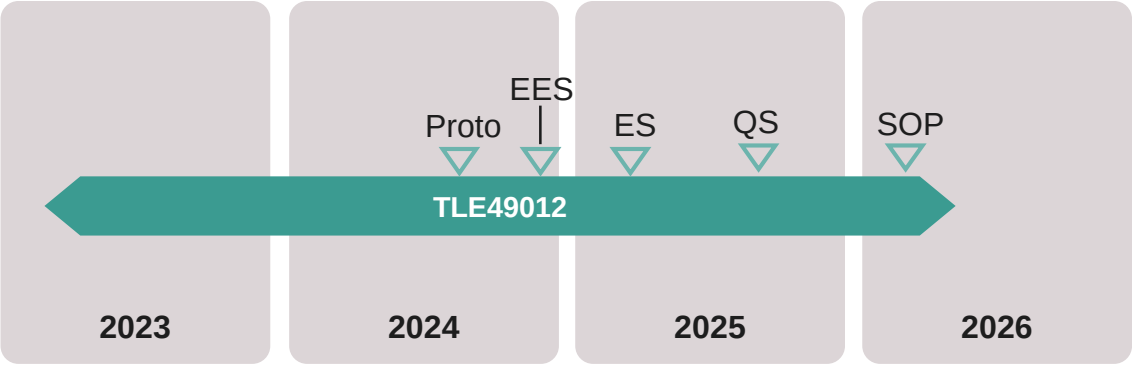
Target Applications

– angular position sensing applications that require high precision and digital interface

Product Highlights

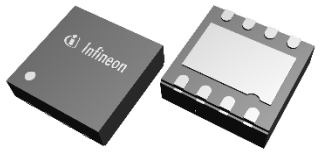
– System self-calibration provides accurate angle measurement without external calibration (<0.8°)
– 3.3V or 5V supply
– small package (3 x 3mm)
– Low latency to support reliable sensing at high dynamics (30 krpm, 25 krad/s²)
– Digital Interface: SPI, ABZ, UVW, PWM

Development Schedule



Note: Development schedule is preliminary and subject to change!

Status



VSON-8

– Product in development
– ES: available
– QS: 09/2025
– SOP: 02/2026

RoHS

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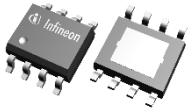
TDA48001
100/80V, 1A Constant Frequency Synchronous Buck Regulator

Features

- Wide 4.2 V to 100 V input range
- Integrated 0.6 or 1 A high-side and low-side switches and bootstrap diode
- Constant frequency current-mode control with 40 ns minimum on-time
- Automatic internal voltage loop compensation
- Selectable forced CCM or high-efficiency PFM operation
- Low quiescent current (< 10 µA) and shutdown current (< 1 µA)
- Frequency settings from 200 kHz to 2.25 MHz
- Optional bypass LDO reduces power loss at high V_{IN}
- Adjustable Undervoltage Lockout (UVLO)/ Enable pin
- On-the-fly V_{OUT} adjust (**QFN only**)
- Synchronizable or optional low-EMI spread spectrum (**QFN only**)
- Thermal shutdown
- Over-current protection at 1.5 A
- VDSON-8, DSO-8 and VQFN-13 with 0.55mm creepage

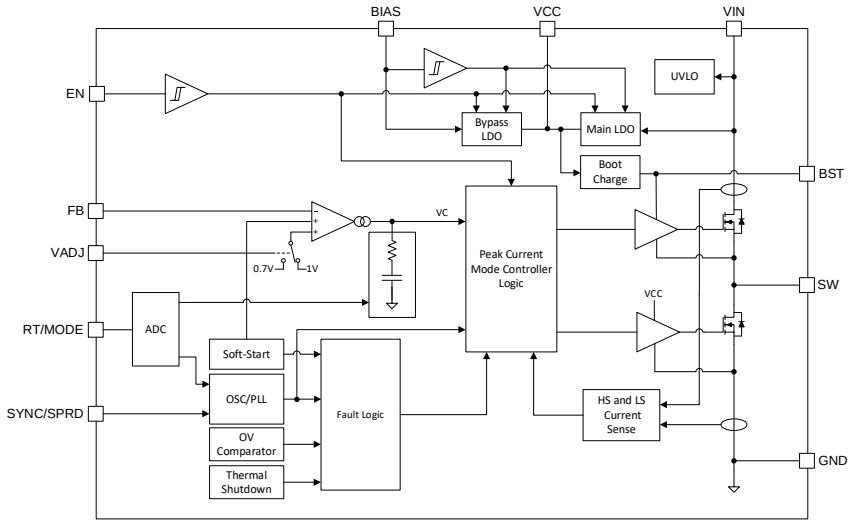
VDSON-8 4x4 mm



DSO-8 5x6 mm

VQFN-13 3x3 mm

Block Diagram



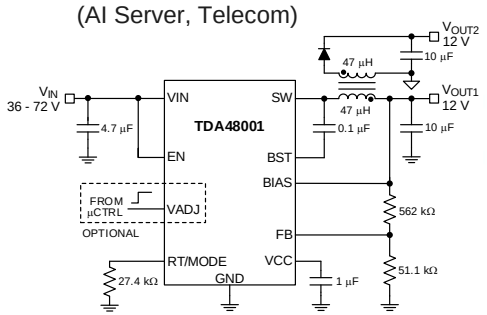
Value Proposition

Fixed-frequency I-mode control	No external RC comp needed – saves space, ease of use
Tiny VQFN-13 package	Small footprint with 0.55 mm creepage – saves space
Efficient high-frequency operation with bypass LDO	Reduces inductor size and higher efficiency for buck apps – saves space and cost
On-the-fly V_{OUT} adjust	Reduces mid-load power loss in telecom/server supplies by modulating gate drive bias – reduces heat/cost/battery consumption
Superior light load efficiency	Extends HV battery pack life by 10-20%
Synch/spread-spectrum	Eliminates EMI concerns – reduces time-to-market

Applications

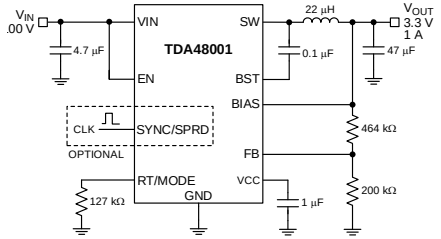
Isolated "Flybuck" bias supply

(AI Server, Telecom)



High-efficiency buck regulator

(LEV, Drones, Robotics, Power Tools, …)



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Focus functional block (2) - Dexterous Hand

Spec & Requirement

- More and more motors (6→12→16→**20+**)
- Power input: 12V, 24V, 48V(?)
- Various tactile sensor solutions while not mature enough
- **Challenge:**
  - Limited size for small motors only while need to carry heavy payload
  - Need to integrate various sensors (e.g. touch, force, temp. sensor)
  - Different robotics hands for different scenarios

Strategy

- Develop & Promote new products with high integration and small size
- Deep-dive sensing technology including:
  - Magnetic sensor: 3D hall sensor (TLV493D)
  - MCU: PSoC touch, PSoC Edge

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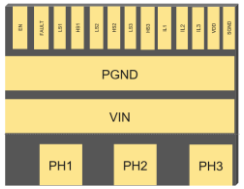
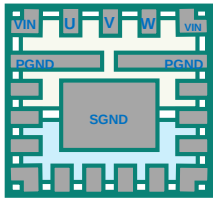
IPM G5 – Three-phase power stage for motor drive application

40, 80 V CoolGaN™ G5 Drive (Target)

RQFN 6x7



RQFN 3.3x3.3

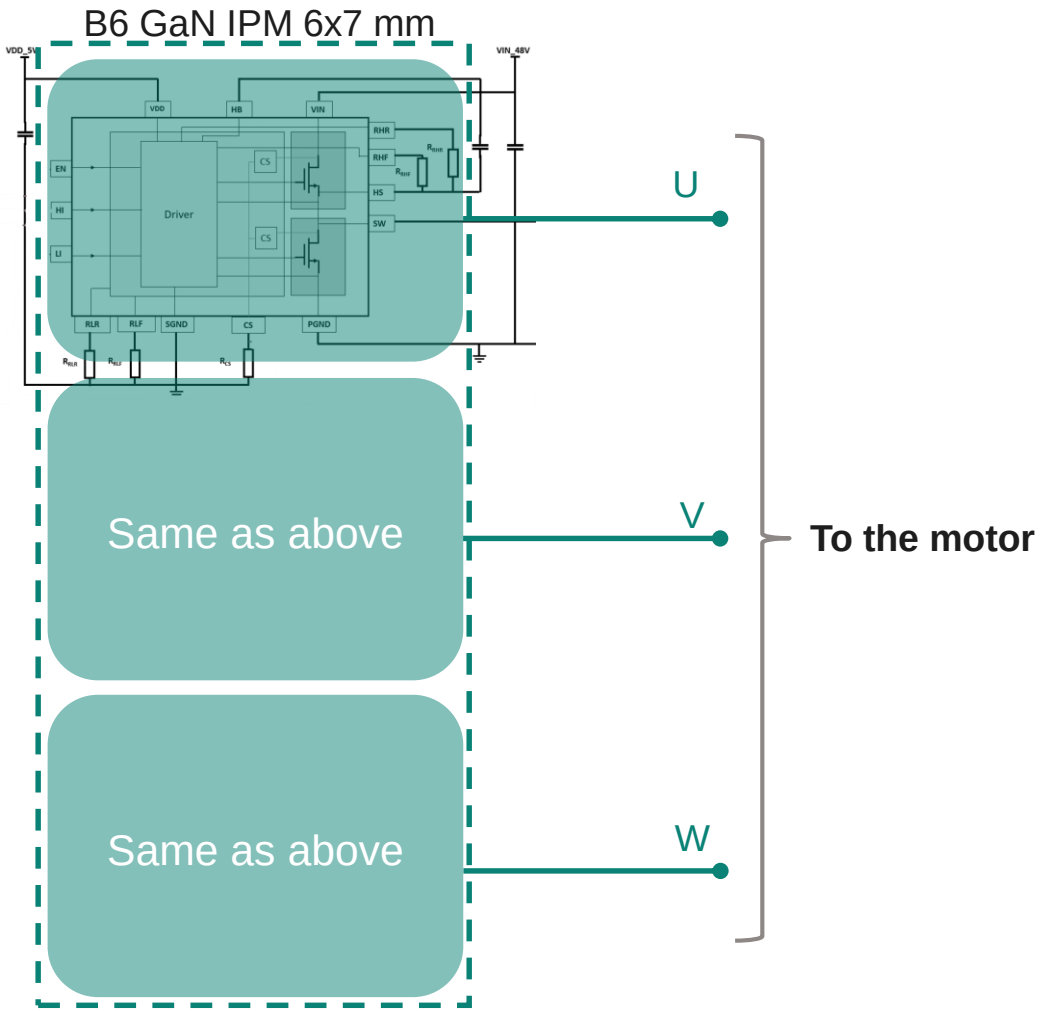


Voltage	$R_{DS(on)}$ typ.	Die GaN size	Package (mm)	Application
80V	6 mΩ	2.1 x 1mm	6 x 7 RQFN DSC	3ph. Motor drive (cobot, joint)
40V	30 mΩ	2 x 1mm	3.3 x 3.3 RQFN DSC	3ph. Motor drive (Dexterous Hands)

Features

- **GaN** based full 3-phase bridge in one package, comprehensive of gate drivers, protections: OVP, UVLO, OCP, CSC, OTP.
- Integrated low side current sensors
- Internal LDO for supply
- Integrated bootstrap capacitors

Applications



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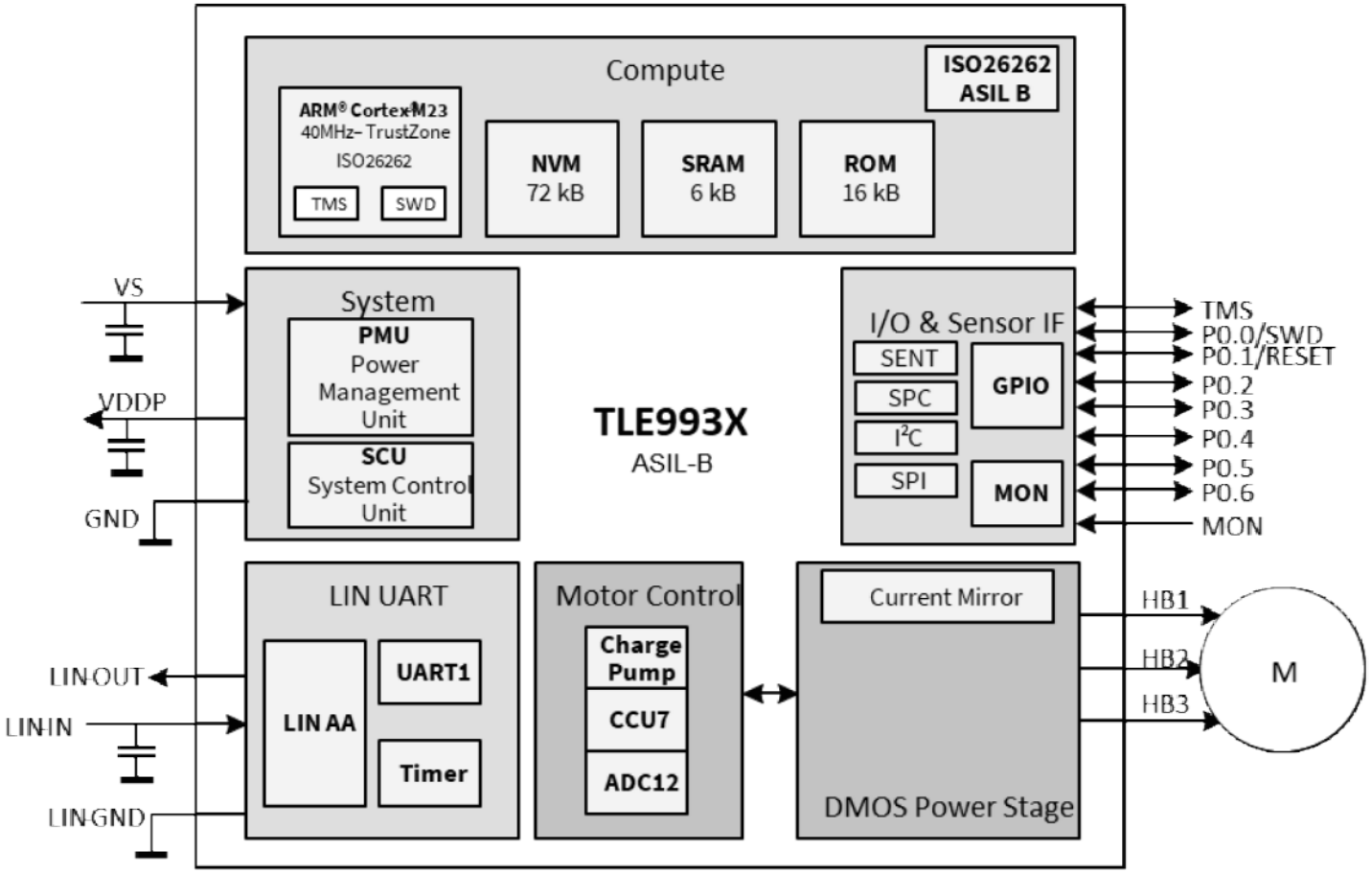
TLE993X
System-on-Chip with Arm® Cortex®-M23

Main Features

- 32-bit Arm® Cortex® -M23 core at up to 40 MHz
- up to 72 kB FLASH
- up to 6 kB RAM and emulated EEPROM
- Fully integrated B6 Bridge capable of 1A RMS / 2A Peak*
- Supply voltage range of 5.5 V to 29 V
- Integrated voltage regulator for external sensor supply
- Integrated diagnosis 12-bit ADC
- Sensor Interface (SENT, I²C, SPI, Analog Inputs)
- TSDSO-18 5x6 mm

*Depending on thermal boundry-conditions



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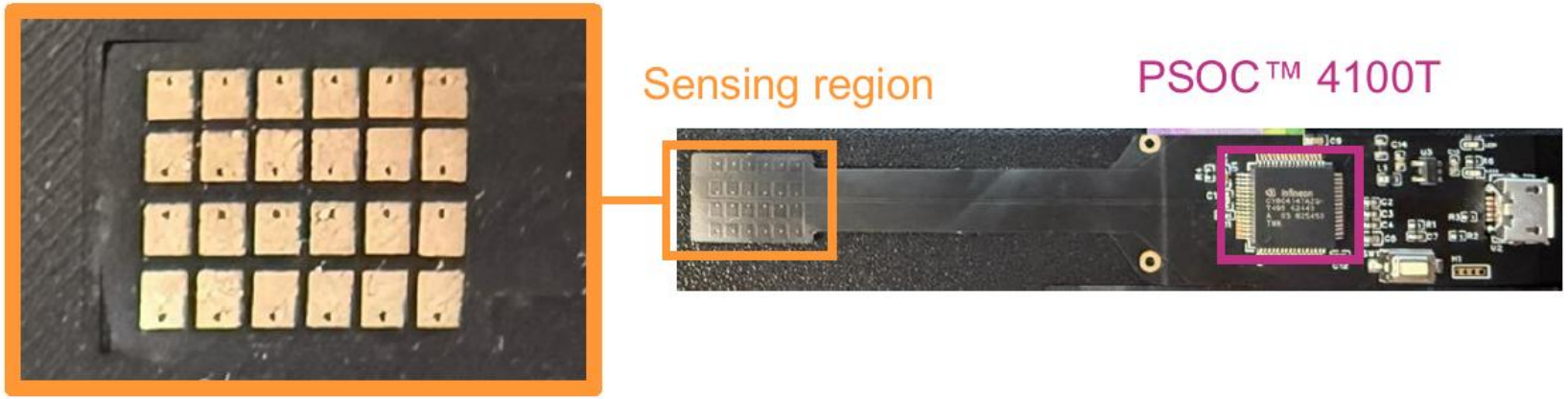
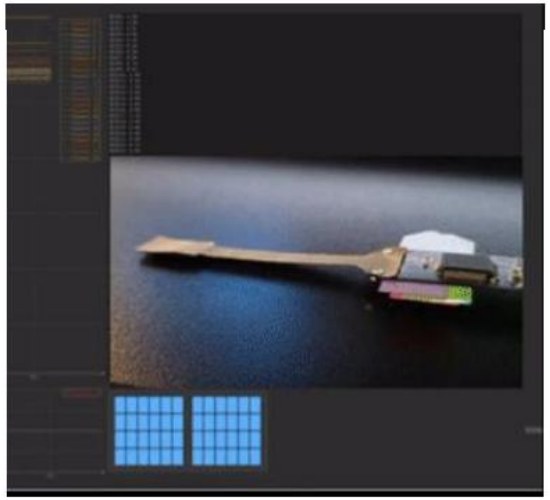
IFX CAPSENSE™ for tactile sensor

- Pressure tactile sensing using CAPSENSE™, utilizing a pressable and flexible sensing frontend

- Key-words:
  - Pressure sensing, tactile sensing, E-skin
  - Multi-channel capacitive sensing
  - 2D force distribution detection, digital touch
  - Dexterous hand





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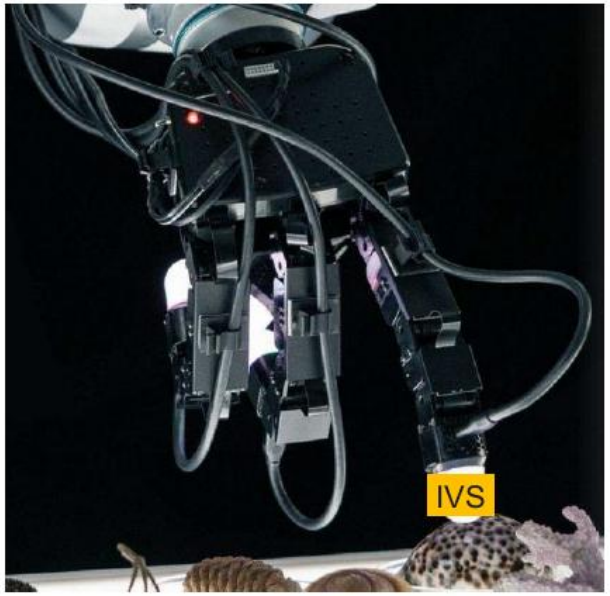
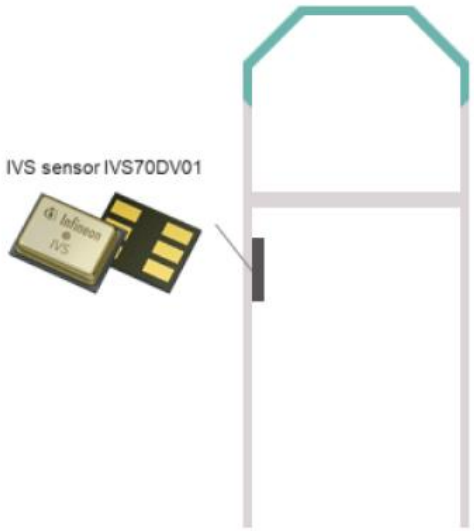
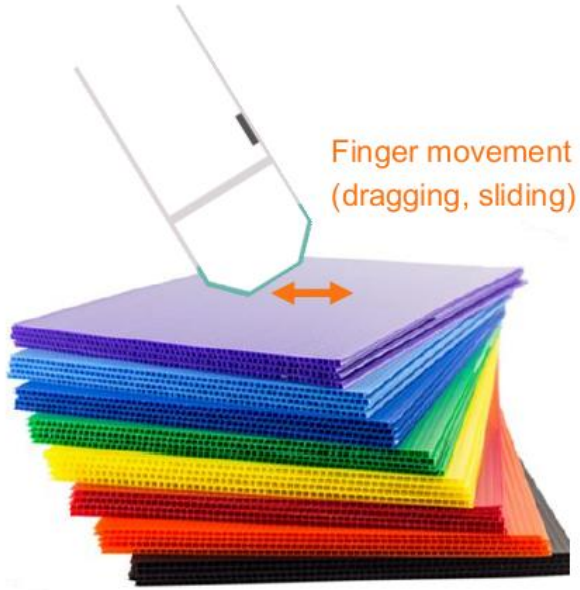
IFX IVS for material classification

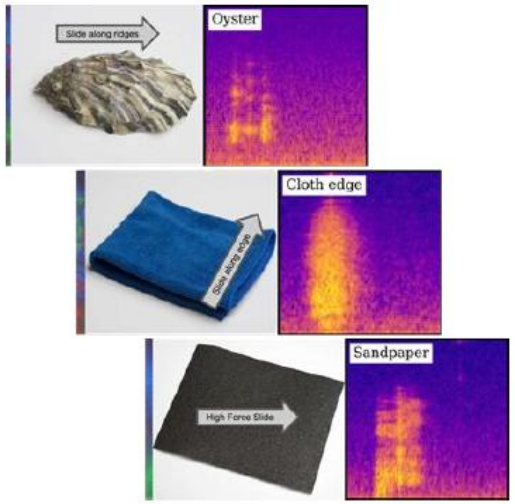
- IVS provides an intuitive measurement for the roughness of an object surface by the surface vibration detected through rubbing, which is similar to how human perceives it
  - Accelerometer can measure similar signal, but IVS has higher SNR

- IVS has suitable bandwidth to detect the surface vibration
  - A normal contact between surfaces has vibration frequency generally from 1k to 5k Hz
  - A sliding between surfaces has fundamental frequency generally around 1k to 2k Hz









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PSOC™ Edge E8X MCU
Enabling a New Generation of Responsive ML Edge Devices

	PSOC™ Edge E81 Base	PSOC™ Edge E83 Base + NPU + Vision	PSOC™ Edge E84 Base + NPU + Vision + GFX
Processor	Cortex-M55 + DSP Cortex-M33	Cortex-M55 + DSP Cortex-M33	Cortex-M55 + DSP Cortex-M33
Machine Learning	DSP NN accelerator, NNLite	DSP NN accelerator, NNLite, U55-128	DSP NN accelerator, NNLite, U55-128
SRAM	Up to 4 MB (System RAM) Up to 1 MB (Low-Power Domain)	Up to 4 MB (System RAM) Up to 1 MB (Low-Power Domain)	Up to 5 MB (System RAM) Up to 1 MB (Low-Power Domain)
RRAM / NVM	512 kB	512 kB	512 kB
External Memory	2x SMIF, 2x SD Host Controller	2x SMIF, 2x SD Host Controller	2x SMIF, 2x SD Host Controller
Audio/Voice	ULP Always ON prog. analog for voice, audio, sensing 4x Analog Mic, 6x Digital Mic Acoustic Activity Detection & NNLite Wake Word	ULP Always ON prog. analog for voice, audio, sensing 4x Analog Mic, 6x Digital Mic Acoustic Activity Detection & U55 ML-based Wake Word & Full Voice Inferencing	ULP Always ON prog. analog for voice, audio, sensing 4x Analog Mic, 6x Digital Mic Acoustic Activity Detection & U55 ML-based Wake Word & Full Voice Inferencing
Graphics	No	No	LP 2.5D GPU Up to 1024 x 768, MIPI-DSI/DBI formats
Vision	No	Position Detection/Face Recognition/Object Detection (VGA) via USB	Position Detection/Face Recognition/Object Detection (VGA) via USB
Peripherals & IO	USB, 10/100 Ethernet, CAN, SPI, UART, I2C, I3C, I2S	USB, 10/100 Ethernet, CAN, SPI, UART, I2C, I3C, I2S	USB, 10/100 Ethernet, CAN, SPI, UART, I2C, I3C, I2S
Security	Secured Enclave, Edge Protect Category 2 and 4	Secured Enclave, Edge Protect Category 2 and 4	Secured Enclave, Edge Protect Category 2 and 4
Packages	BGA-220, 10x10, 0.65p WLB-154, 4.3x5.3, 0.35p	BGA-220, 10x10, 0.65p WLB-154, 4.3x5.3, 0.35p	BGA-220, 10x10, 0.65p eWLB-235, 7x7, 0.4p WLB-154, 4.3x5.3, 0.35p
Availability	Samples – Q2-2025 Production – Q2-2025	Samples – Q2-2025 Production – Q2-2025	Samples – Now Production – Q2-2025

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Imagimob with ModusToolbox™ from data to edge model

Build a model

Bring your own model

OR

Buy a Ready Model (no optimization needed)

Optimize & validate your model

Develop your embedded product with IFX MCUs

Imagimob Studio

Data Acquisition → Preprocessing → Model Training

- Data Import
- Data Capture
- Data Generation
- Data Augmentation
- Data Labeling
- Preprocessor / Feature extractor selection
- Model Selection
- Model Training
- Model Validation

Imagimob Studio + ModusToolbox™

Optimization

- Hardware Optimization

ModusToolbox™

.c .a

Deployment

- Code Generation
- Target Device Test and Validation
- Field Deployment

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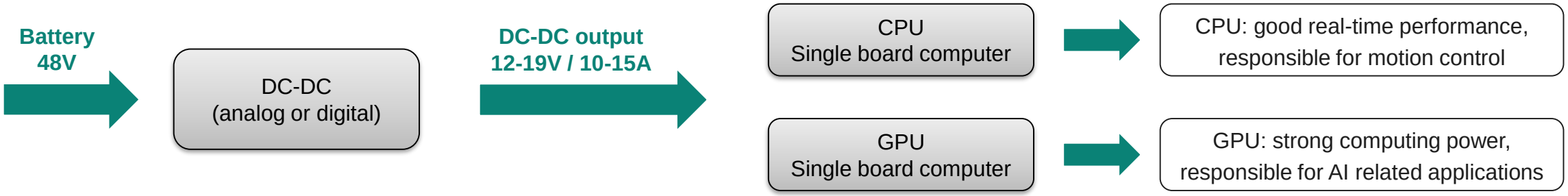
Focus functional block (3) - Power Management

Power flow for Humanoid Robot's Main Compute



Spec & Requirement

- Main Compute (CPU & GPU) boards are outsourcing and compatible with wide range input voltage (12-19V)
- Humanoid Robot customers develop the DC-DC by themselves (Buck, power for CPU, GPU and others)
- Vairous solutions in the market from different customers
- MV GaN is being evaluated as SR in DC-DC

Strategy

- Secure IFX share for multiphase controller & DrMOS at core board decision maker:
  - CPU: Intel
  - GPU: NVIDIA & Horizon Robotics
- Closely follow up power requirement for new NVIDIA Orin series GPU (e.g. Thor)
- Develop IFX DC-DC integrated solution with better system performance (efficiency & size)

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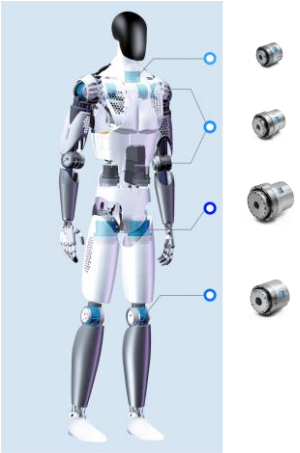
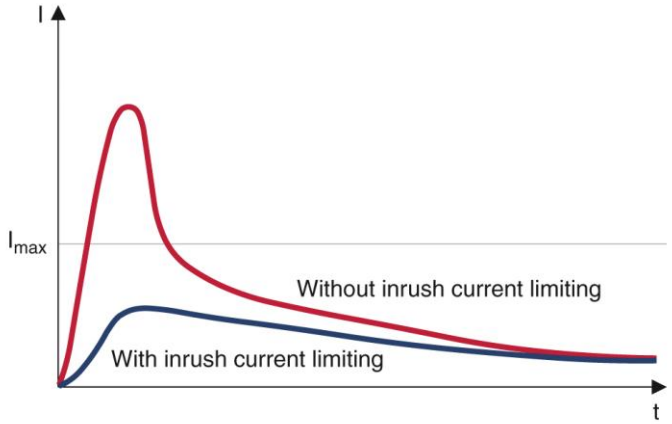
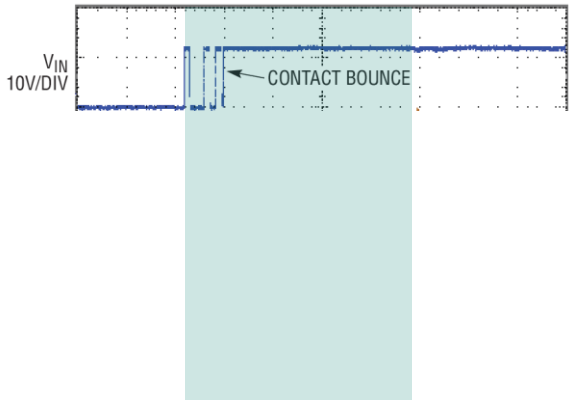
HotSwap application in Humanoid Robot

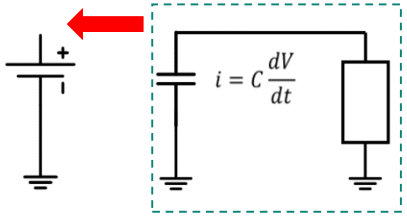
– What is Hot-Swap?



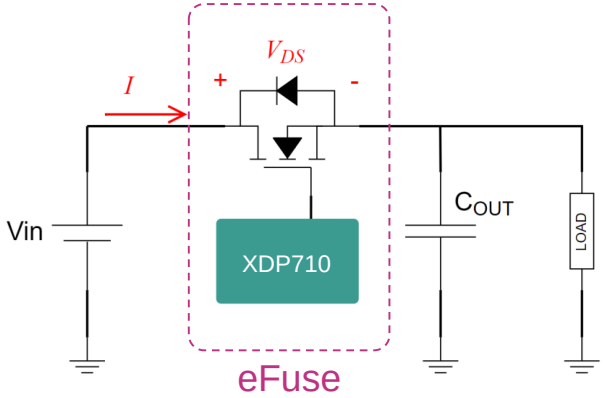


– Hot-Swap issue





– Hot-Swap controller/E-fuse



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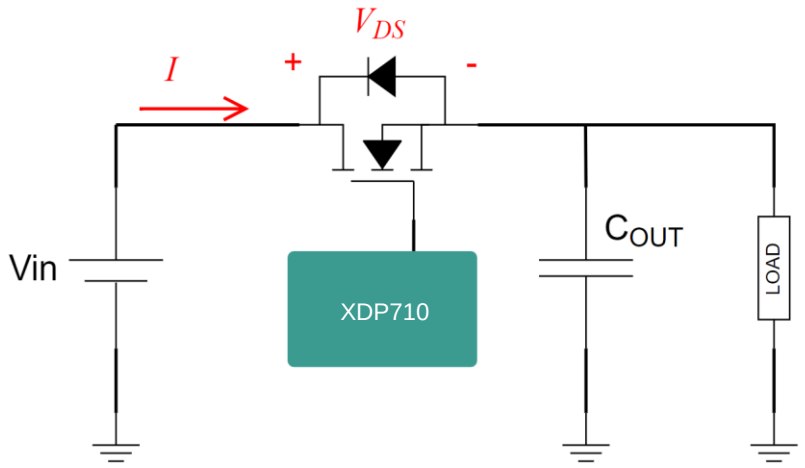
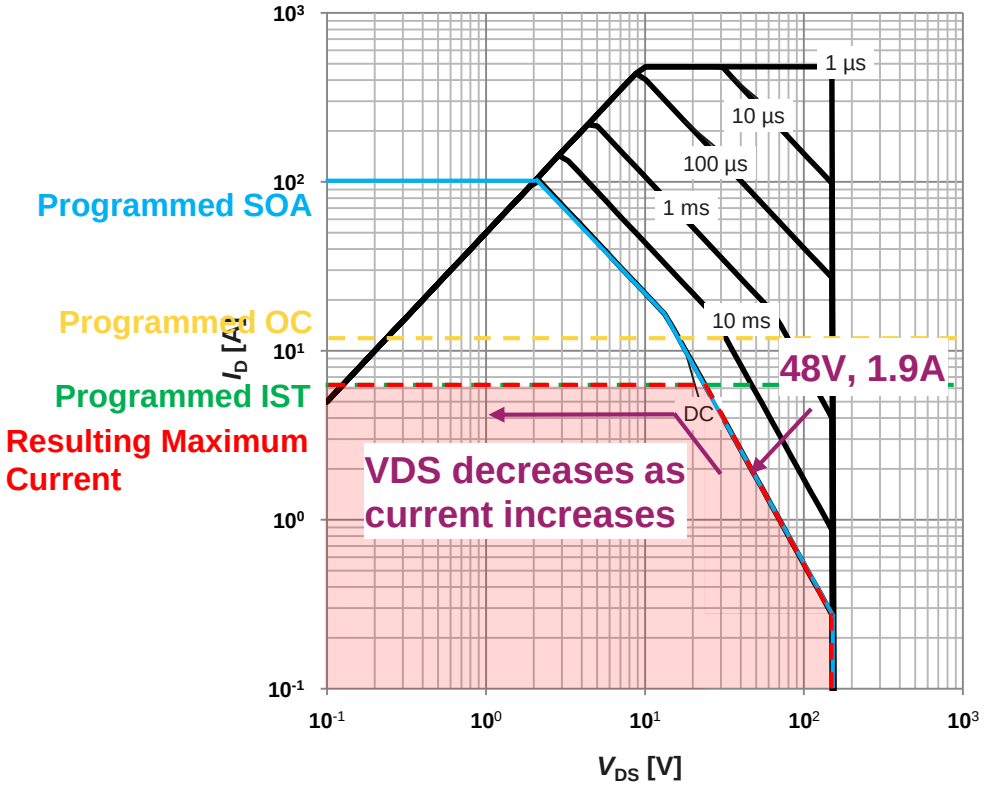
XDP710-002
Hot-swap controller for Humanoid Robot

– Our approach: “Digital“ SOA control of FET



– Advantages:
  – Safest solution – FET and Load are protected
  – Pre-programmed SOA for Infineon FETs
  – Fully optimized
  – Cost competitve
  – Fully Digital mode of operation (low cost BOM)
  – Analog-Assisted mode of operation (for backward compatability)
  – Any FET‘s SOA can be programmed using GUI



IPB072N15N3 SOA

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1EDL8011
High side Gate Driver for Humanoid Robot

Description

The 1EDL8011 gate driver IC provides **fast turn-on and turn-off** of **high side** N-channel MOSFETs with its powerful gate current capabilities. It consists of **an integrated charge pump** with an external capacitor to provide strong start-up. The gate driver IC is used to manage inrush current and protects in case of faults. Undervoltage Lockout (**UVLO**) protection at input voltage prevents the device from operating under hazardous conditions. The 1EDL8011 is available in DSO-8 package including **OCP protection** feature, an **adjustable current setting threshold**, time delay and a safe start-up mechanism with flexible blanking during MOSFET turn-on transitions.

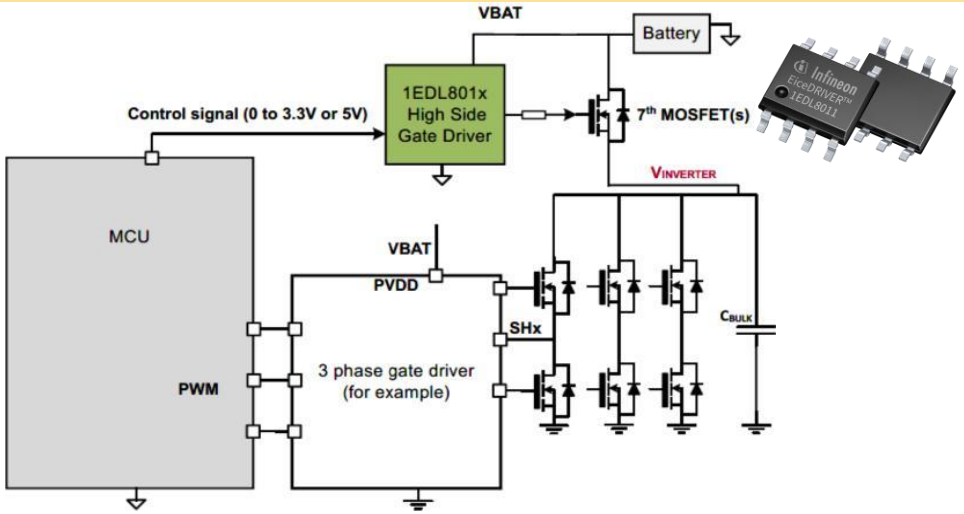
Key features

1	Integrated **125V** high side safety disconnect switch gate driver
2	Strong gate drive and charge pump for **fast turn-on/off**
3	Over-Current protection (**V_{DSON} Sense**)
4	V_{GS} safe switch off mechanism
5	**Adjustable** current trip **threshold**

Use cases and target applications

- Power tools
- Garden tools
- Robots and vacuum cleaners
- E-bikes
- Drones, multi-copters

System / application infrastructure



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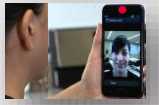
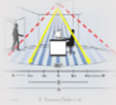
Focus functional block (4) - Environmental Sensor

Environmental	Pressure	3D Radar	3D iToF	Microphone & ultrasound TR
6x6mm² World smallest form factor	Best-in-class resolution	Application innovation leader	 Power efficient Indirect Time of Flight 3D camera	Unique SNR, sensitivity and power efficiency
CO_2 Best accuracy	Energy Efficiency	 Best performance & energy efficiency	High resolution 3D point cloud	Crystal clear audio signals

Use cases in robotics

- Collision avoidance / obstacle-stairs detection (radar,ToF,ultrasounds)
- Improved User Interface & experience (gesture, VUI)
- Height/floor positioning
- Odometry fusion / positioning / SLAM
- Vacuum flow mgt
- Material classification
- Human Presence/safety

Requires sensor+MPU/MCU+algo/software integrated system offer

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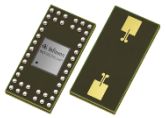
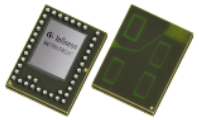
Wide offering of different and advanced Navigation & HMI Sensing

RADAR


BGT60LTR11AIP


BGT60TR13C

BGT60ATR24C

Features

- Small size including Antenna in Package
- Accurate measurement down to 1cm distance
- Targets detection up to 15m (front facing)
- Range resolution to ~3cm
- Sub-mm movements detection
- Power consumption <5mW

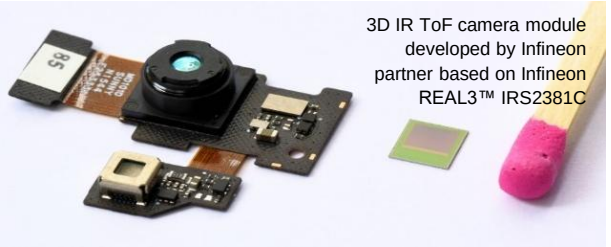
Use cases

3D IR ToF Imaging

- ✓ Obstacle avoidance
- ✓ Small objects recognition
- ✓ Simultaneous Localization & Mapping

Features

- Robust and accurate depth data
- Full operation in bright sunlight
- Fast one-time calibration
- Mechanically robust w/o moving parts
- Low depth-map computation efforts


3D IR ToF camera module developed by Infineon partner based on Infineon REAL3™ IRS2381C

MEMS Microphone

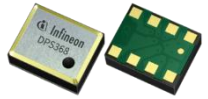
- ✓ Sound detection for HMI
- ✓ Objects detection for navigation

Features

- Ultra high SNR up to 73dB
- Ultra high Accoustic Overload Point up to 135dBSPL
- Sensitivity up to -38dBV & Omnidirectional directivity

Air Pressure Sensor

- ✓ Height / altitude sensing
- ✓ Air flow control
- ✓ Liquid level monitoring



Features

- Pressure range 0.3 – 1.2bar
- Precision up to ±0.002mbar or ±2cm
- Abs. accuracy ±1mbar or ±8m
- IPx8 certified if necessary
- Integrated temperature sensor

3D Magnetic Sensor with integrated Temperature Sensor

- ✓ Joysticks for HMI
- ✓ Pushbuttons for HMI
- ✓ Highly precise position control

Features

- 3D magnetic flux density sensing up to ±160mT
- Programmable sensitivity up to typ. 30.8 LSB_{12}/mT

CO_2 Sensor

Features

- 4x smaller & 3x lighter than non-dispersive IR sensor counterpart
- Sensing range up to 32000ppm
- Accuracy ±30ppm ± 3% within 400 – 5000ppm range

Cap Sense

Features

- Best-in-class noise immunity
- Improved EMC, CMR, and DC rejection
- Hardware-based wake-on-touch

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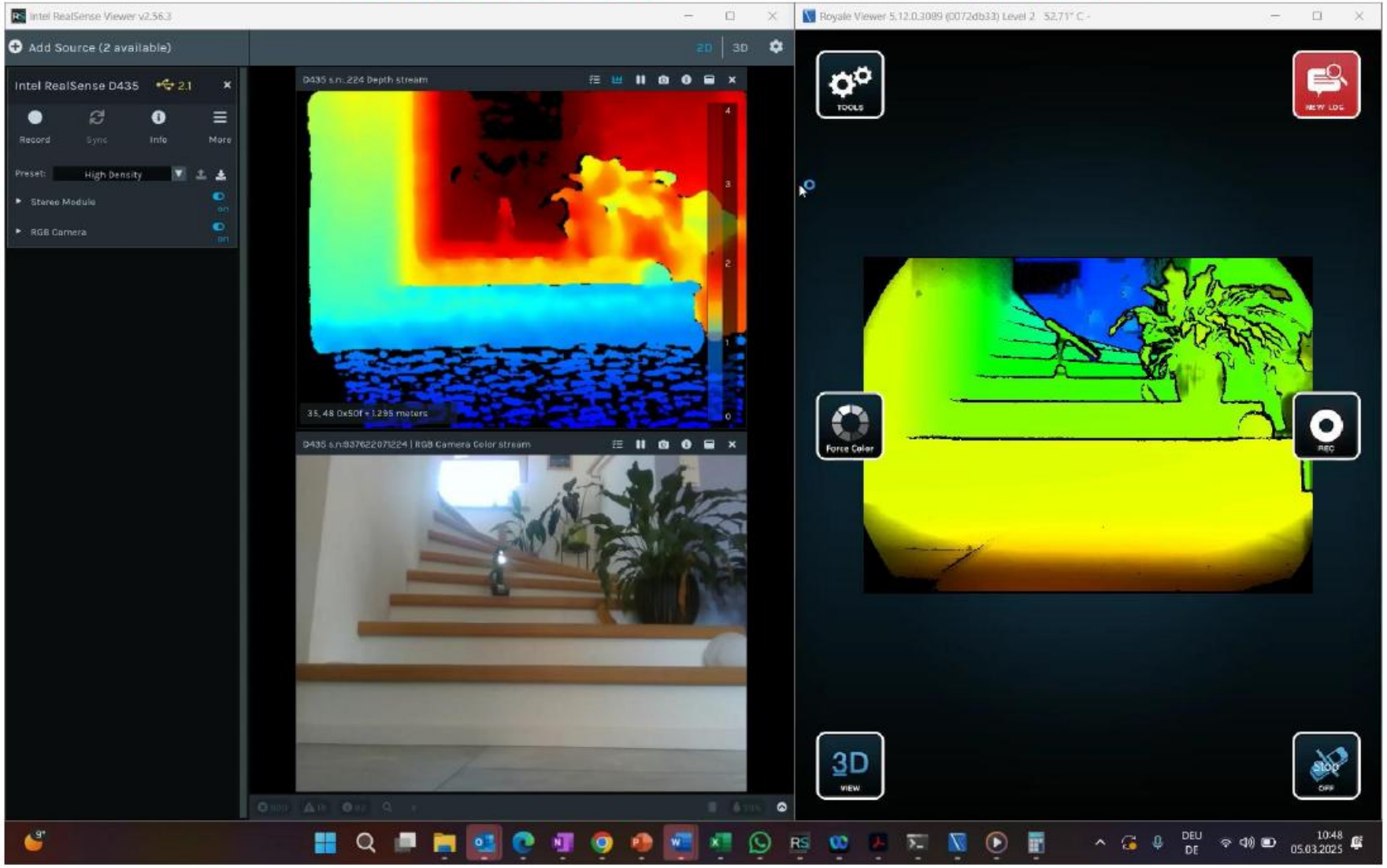
RealSense Vs IFX ToF

RealSense D435

IRS2976C



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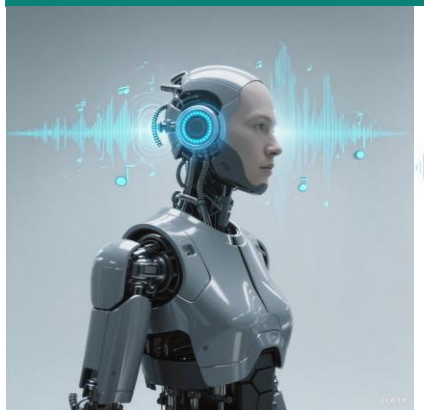
Human & Robot Microphones Voice Interaction

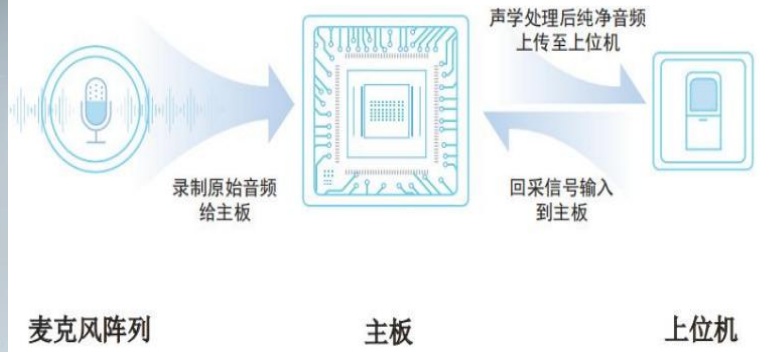
Solution Description

ELEVOC's humanoid robotics hearing solution, features include:

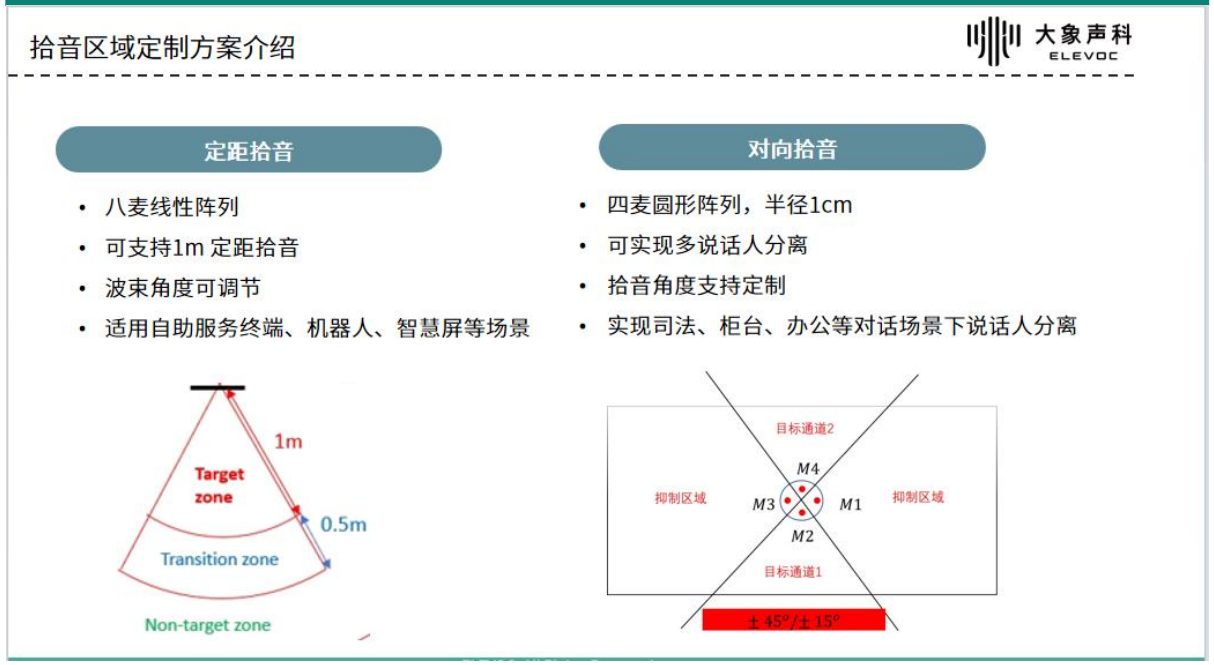
- AI noise reduction: suppress noise to improving the voice recognition rate.
- Echo Cancellation: Supports 2-ch HW loopback access to eliminate echo from the local speakers
- Sound source location : Real-time sound source localization function with a positioning accuracy of ±5 degrees.
- Voice Wake-up: Wake-up words is customizable, low-power mode and oneshot
- Customized Mic-array design: linear & circular array, 1/2/4/8x mics optional
- **Customized Sound Pickup Area:** 360° omnidirectional sound pickup, fixed-angle sound pickup, and fixed-distance sound pickup.

Humanoid Robotics "Hearing" System



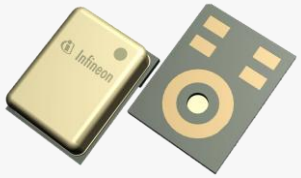


Key Feature—Customized Sound Pickup Area



Included Infineon products

- IM72D128 MEMS microphone



- AOP 128dBSPL
- SNR 72dB(A)
- Sen. -36dBFS
- Interfaces Digital PDM
- Size 4*3*1.2mm³

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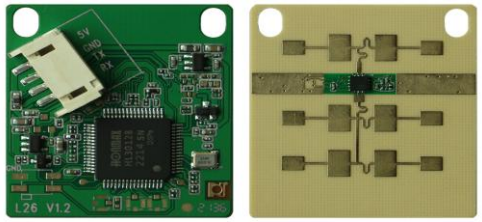
Human & Robot Radar Perception Interaction

Gesture Detection

- **XENSIV™ 60GHz 1T3R Radar BGT60TR13C**
- Human & Robot contactless interaction
- Range:: 0.3-1m (gesture), 3m (presence)
- Gesture type: up, down, left, right, tap
- Recognition trigger time: ≤0.5s
- Recognition rate: ≥95%

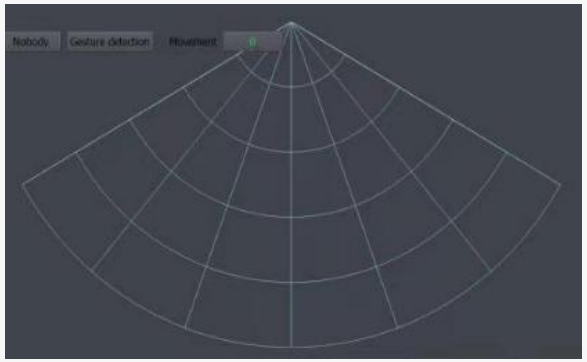
Human Presence Detection

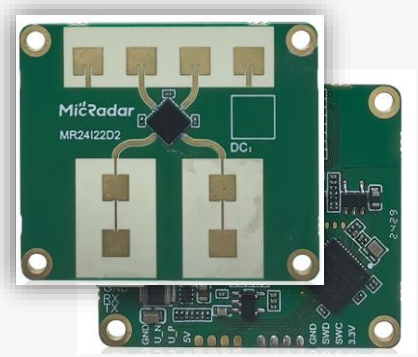
- **XENSIV™ 24GHz 2T2R Radar BGT24LTR11N16**
- Wake up robot for low power consumption
- 1-4m human statics detection
- Angle & range can be configured



People Tacking & Segmentation

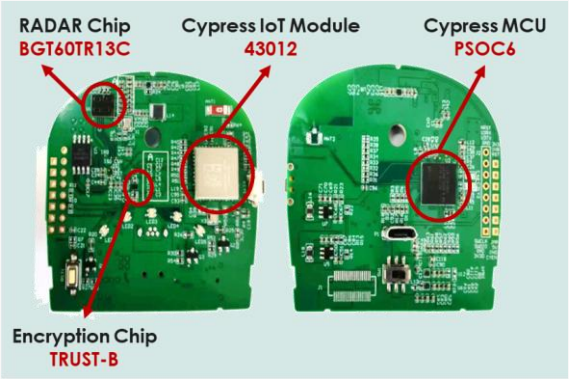
- **XENSIV™24GHz 2T2R Radar BGT24LTR22**
- identify human intentions & reduce false recognition
- Distance range 1-5m, with 0.3m accuracy
- FOV H-100°, E-80°, and angular resolution ≤10°





Human Vital Sensing

- **XENSIV™ 60GHz 1T3R Radar BGT60TR13C**
- MCU PSOC CY8C6245, WIFI/BT CYW43012, OPTIGA ™ Trust B
- Continuously monitor human respiration and heartbeat vital signs by detecting the chest movement by radar
- **Elderly Care Robot**



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Focus functional block (5) - Connectivity (Wi-Fi & Bluetooth)

Wi-Fi 6		Wi-Fi 6E			Wi-Fi 6E SoC
2x2 MIMO	1x1 SISO	2x2 MIMO	1x1 SISO	1x1 SISO	1x1 SISO
80MHz	80MHz	80MHz	80MHz	20MHz	20MHz
Dual-band 2.4/5 GHz	Dual-band 2.4/5 GHz	Tri-band 2.4/5/6 GHz	Tri-band 2.4/5/6 GHz	Tri-band 2.4/5/6 GHz	Tri-band 2.4/5/6 GHz
Max. 1200Mbps	Max. 600Mbps	Max. 1200Mbps	Max. 600Mbps	Max. 143Mbps	Max. 143Mbps
PCIe, SDIO	PCIe, SDIO	PCIe, SDIO	PCIe, SDIO	SDIO, GSPI	SDIO, GSPI
Bluetooth 5.3	Bluetooth 5.3	Bluetooth 5.3	Bluetooth 5.3	Bluetooth 5.3(LE Audio)	Bluetooth 5.3
FCBGA, 0.65mm pitch M.2 Module WLCSP, 0.2 mm pitch SiP Module WLBGA, 0.35 mm pitch small form factor		FCBGA, 0.65mm pitch M.2 Module WLCSP, 0.2 mm pitch SiP Module WLBGA, 0.35 mm pitch small form factor		WLBGA, 0.35mm pitch Small form factor	WLBGA, 0.35mm pitch Small form factor

- **Integrated with iPA, iLNA, PMU to save BOM cost**
- **Small form factor**
- **Three products tailored to a range of applications**
- **Footprint compatible in the product family**

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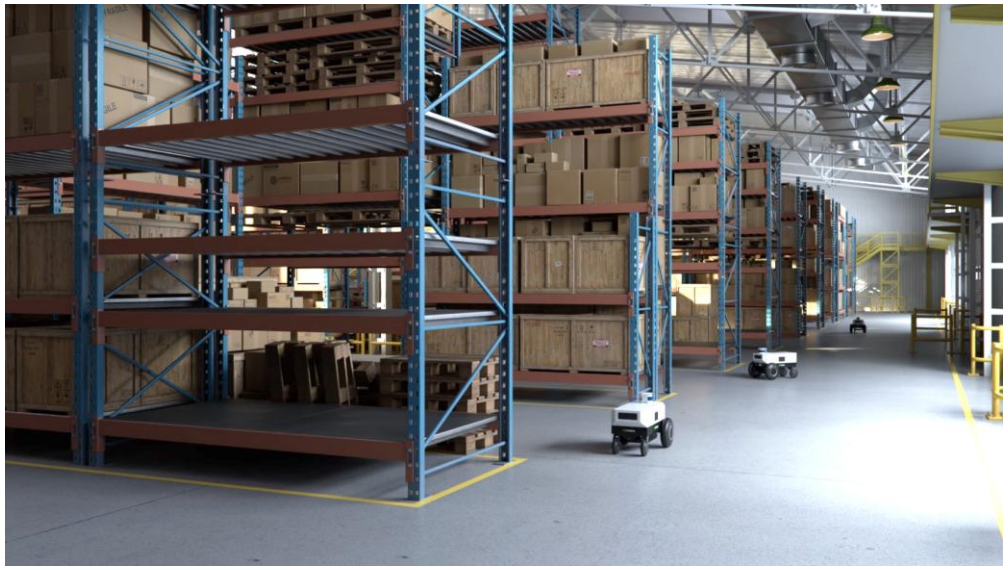
Example: Fleet Management for AMRs Requires Extended Range

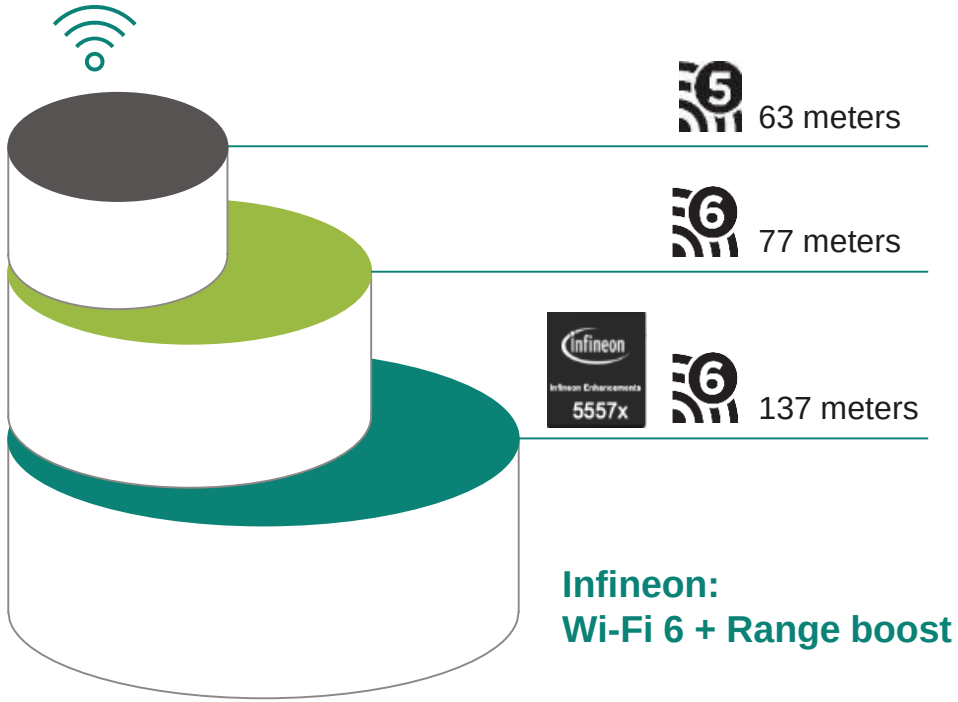


Challenges:

- Robots operate in large warehouses
- Communications go through obstacles causing weak signals
- Robots lose connections/ communications
- Efficient power to extend service time

Infineon Wi-Fi 6 + Range boost

- 2x range improvement vs. Wi-Fi 5
- >70% improvement from Wi-Fi 6



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